Department of
Computer Science and Engineering

University of Dhaka

Title: Implementation of Numerical Solution of Ordinary Differential Equations (ODEs)

CSE 3212: Numerical Analysis Lab

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1 Introduction

Ordinary differential equations (ODEs) appear in physics, engineering, biology, economics, and many other scientific and applied fields. In most real-world situations, ODEs cannot be solved analytically, and therefore numerical methods are required.

This lab focuses on two of the most widely used numerical techniques for solving first-order initial value problems (IVPs):

- **Euler's Method** – the simplest numerical scheme, first-order accurate.
- **Heun's Method (Improved Euler Method)** – a predictor–corrector scheme, second-order accurate.
- **Mid Point Method (Improved Euler Method)** – a predictor–corrector scheme, second-order accurate.

Students will learn how to implement these methods step-by-step and compute approximate solutions to initial value problems.

2 Objectives

After completing this lab, students will be able to:

- Understand initial value problems and numerical integration of ODEs.
- Apply Euler's Method to numerically solve first-order ODEs.
- Apply Heun's and Midpoint Method (Improved Euler Methods) for improved accuracy.
- Estimate local and global truncation errors.
- Write and implement algorithmic steps using Python.

3 Background Theory

3.1 Ordinary Differential Equations (ODEs)

Many real-world systems in physics, biology, engineering, and economics are governed by differential equations. These equations describe how a quantity changes with respect to an independent variable (usually time).

A first-order ODE has the form:

$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0$$

where $y(t)$ is the unknown function, $f(t, y)$ defines the rate of change, and $y(t_0) = y_0$ is the initial condition.

However, most ODEs cannot be solved analytically. Therefore, numerical methods are required to approximate the solution at discrete time points.

3.2 Euler's Method

3.2.1 Concept and Motivation

Euler's Method is the simplest numerical technique for solving first-order ODEs. It is based on approximating the solution curve using the tangent line at each step.

Given:

$$\frac{dy}{dt} = f(t, y),$$

the Euler update formula is:

$$y_{n+1} = y_n + hf(t_n, y_n),$$

where h is the step size.

3.2.2 Geometric Interpretation

At each point (t_n, y_n):

1. Compute the slope $m = f(t_n, y_n)$.
2. Move horizontally by h.
3. Move vertically by $h \cdot m$.

This constructs a polygonal approximation to the true solution.

3.2.3 Limitations

- Low accuracy unless a very small step size is used.
- Highly sensitive to stiffness.
- May overshoot or undershoot the true solution curve.

3.3 Heun's Method (Improved Euler / Modified Euler Method)

3.3.1 Motivation

Heun's Method improves Euler's Method by using an average of two slopes. This makes it more accurate and more stable. It is also known as the Improved Euler Method or a form of the Runge–Kutta 2nd order method (RK2).

3.3.2 Concept

Heun's method uses two slopes:

- Predictor slope (Euler's slope):
$$k_1 = f(t_n, y_n)$$
- Corrector slope:
$$y_{\text{pred}} = y_n + hk_1,$$
$$k_2 = f(t_n + h, y_{\text{pred}})$$

Average slope:
$$\bar{k} = \frac{k_1 + k_2}{2}$$

Final update:
$$y_{n+1} = y_n + h\bar{k}$$

3.3.3 Geometric Interpretation

1. Euler predicts a point using the initial slope.
2. Heun corrects this prediction using the slope at the predicted point.
3. The final slope is the average of the two.

3.3.4 Advantages

- Much more accurate than Euler's Method.
- More stable for many systems.
- Reasonable computational cost.

3.3.5 Limitations

- Requires two function evaluations per step.
- Not as accurate as higher-order Runge–Kutta methods (e.g., RK4).

Feature	Euler Method	Heun Method
Order of Accuracy	First order	Second order
Slopes Used	1 (initial)	2 (initial + predicted)
Stability	Poor	Better
Computational Cost	Low	Moderate
Accuracy	Low	High

3.4 Comparison Between Euler and Heun Methods

Heun's Method is generally preferred for practical engineering and scientific applications unless extremely fast computation is required.

4 Illustrative Example (Euler & Heun)

Problem: Solve the IVP

$$\frac{dy}{dx} = x^2 - y, \qquad y(0) = 1$$

Compute approximate values at $x = 0.1,\ 0.2,\ 0.3$ using $h = 0.1$.

4.1 Euler Method Solution

Step 1: At $(x_0, y_0) = (0, 1)$,

$$f(0,1) = 0 - 1 = -1$$

$$y_1 = 1 + 0.1(-1) = 0.9$$

Step 2: At $(x_1, y_1) = (0.1, 0.9)$,

$$f(0.1, 0.9) = 0.01 - 0.9 = -0.89$$

$$y_2 = 0.9 + 0.1(-0.89) = 0.811$$

Step 3: At $(x_2, y_2) = (0.2, 0.811)$,

$$f(0.2, 0.811) = 0.04 - 0.811 = -0.771$$

$$y_3 = 0.811 + 0.1(-0.771) = 0.7339$$

4.2 Heun's Method Solution

Step 1: At $(x_0, y_0) = (0, 1)$,
Predictor:

$$y_p = 1 + 0.1(-1) = 0.9$$

Compute slopes:

$$f_1 = -1, \qquad f_2 = f(0.1, 0.9) = -0.89$$

Corrector:

$$y_1 = 1 + \frac{0.1}{2}(-1 - 0.89) = 1 - 0.0945 = 0.9055$$

5 Lab Tasks (Please implement yourself and show the output to the instructor)

5.1 Lab Task 1

Solve the following initial value problem (IVP) over the interval

$$0 \leq t \leq 5, \qquad y(0) = 1,$$

where the differential equation is

$$\frac{dy}{dt} = (1 + 4t) - y.$$

and the exact solution of the above differential equation is

$$\boxed{y(t) = 4t - 3 + 4e^{-t}}$$

Compute numerical solution using:

(a) Euler's Method.

(b) Heun's Method.

(c) Mid Point Method.

Perform computations for five different step sizes:

$$h = 0.50, \qquad h = 0.25, \qquad h = 0.1, \qquad h = 0.01 \qquad h = 0.001.$$

For each method and each step size:

- Generate a complete table of t_n, numerical y_n, exact $y(t_n)$, and absolute error.
- Plot all results (Exact, Euler, Heun, Midpoint) on the same graph.

Sample Tables for Step Size $h = 0.25$

Exact Solution Table ($h = 0.25$)

t	$y(t) = 4t - 3 + 4e^{-t}$
0.00	1.000000
0.25	4(0.25)-3+4e$^{-0.25}$ = 0.213061
0.50	-1+4e$^{-0.5}$ = 0.426122
0.75	0 + 4e$^{-0.75}$ = 0.944162
1.00	1 + 4e^{-1} = 2.471516

(Students must compute all values up to $t = 5$.)

Euler Method Results Table ($h = 0.25$)

t_n	y_n (Euler)	$y(t_n)$ exact	Error $\lvert y - y_n\rvert$
0.00	1.000000	1.000000	0.000000
0.25	1 + 0.25[(1+0)-1] = 1.000000	0.213061	0.786939
0.50	...	0.426122	...
0.75	...	0.944162	...
1.00	...	2.471516	...

(Students complete up to $t = 5$.)

Heun Method Results Table ($h = 0.25$)

t_n	y_n (Heun)	$y(t_n)$ exact	Error $\lvert y - y_n\rvert$
0.00	1.000000	1.000000	0.000000
0.25	students compute	0.213061	...
0.50	...	0.426122	...
0.75	...	0.944162	...
1.00	...	2.471516	...

(Students repeat process up to $t = 5$.)

Tables for Step Size $h = 0.50$

Exact Solution Table ($h = 0.5$)

t	$y(t) = 4t - 3 + 4e^{-t}$
0.0	1.000000
0.5	0.426122
1.0	2.471516
1.5	3.903958
2.0	4.459697

(Students compute up to $t = 5$.)

Euler Method Table ($h = 0.5$)

t_n	y_n (Euler)	Exact $y(t_n)$	Error
0.0	1.000000	1.000000	0.000000
0.5	$\mathrm{y}_1 = y_0 + 0.5[(1+0) - 1] = 1.0000$	0.426122	0.573878
1.0	...	2.471516	...
1.5	...	3.903958	...
2.0	...	4.459697	...

Heun Method Table ($h = 0.5$)

t_n	y_n (Heun)	Exact $y(t_n)$	Error
0.0	1.000000	1.000000	0.000000
0.5	...	0.426122	...
1.0	...	2.471516	...
1.5	...	3.903958	...
2.0	...	4.459697	...

Graphing Requirement

Students must produce a graph with:

- Exact solution curve $y(t)$
- Euler approximations (With different step sizes)
- Heun approximations (With different step sizes)
- Midpoint approximations (With different step sizes)

All results must be displayed in a **single graph** to clearly compare accuracy and convergence behavior.

5.2 Lab Task 2

Solve the initial value problem (IVP)

$$\frac{dy}{dt} = y \sin^3(t), \qquad y(0) = 1,$$

over the interval

$$0 \leq t \leq 5$$

using a step size

$$h = 0.1 \qquad h = 0.005 \qquad h = 0.001 \qquad h = 0.0005 \qquad h = 0.0001$$

The exact solution of the above IVP is,

$$\boxed{y(t) = \exp\left[-\cos t + \frac{\cos^3 t}{3} + \frac{2}{3}\right]}$$

Compute numerical solution using:

(a) Euler's Method.

(b) Heun's Method.

(c) Midpoint Method.

For each method:

- Create a complete table of t_n, numerical y_n, exact $y(t_n)$, and absolute error.
- Plot all solutions (exact, Euler, Heun and Midpoint) on the same graph.